Audit of *Safe prompts, safe fields*

the order-1 Wasserstein conformal region for linear flow matching

flow_matching_safe_set_POC/safe_prompt_fields.tex, version of 14 October 2026 (29 pages)

Mathematical and experimental audit

14 October 2026

Abstract

Every numbered statement of the note (Sections 2–7 and Appendix A) was re-derived line by line, every script of the two experiments of Sections 9 and 10 (code/e1_*.py, code/e2_*.py, code/make_reflow_figure.py) was read against the text, and the numbers that carry a conclusion were recomputed from results/. Three new checks were run on the same pools and models, without refitting anything: a probe-sensitivity analysis, a second attack that searches the directions a region penalises least, and a Mahalanobis prompt-space baseline (code/e1_robustness.py, code/e2_robustness.py; every number below is produced by code/audit_checks.py).

Headline. The theory is sound: every theorem, proposition and lemma is correct as stated, and its proof is complete. The defects are in the experiments and in the prose that draws conclusions from them. Three experimental conclusions do not survive re-checking. (F1) The finding that L^2 and L^p map distances are less sensitive than the truncated L^1 distance is produced by *one* of the 256 probe noises, which the 1-rectified sampler sends far from the data; without it, L^2 is at least as sensitive, as Proposition 7.3 predicts. (F2) The pull-back region of the 1-rectified model is broken by an attack that searches the low-eigenvalue directions of its metric ($P(A) = 0.22$ instead of the reported 0.030). (F3) A Mahalanobis prompt-space region excludes every unsafe request and resists the note's attack as well as any field region, because the approved embeddings of the toy are degenerate; the note's "prompt versus field" conclusion is shown only against Euclidean prompt distances.

Other defects. Two statements in the prose are false (convexity of the improved region, a "Markov-free" certificate) and three overstated (the size of the Grönwall factor, the mechanism attributed to the floor of Proposition 5.6, a comparison of sensitivities), ten reporting or methodology defects (among them a misreported coverage range, a certificate check that cannot fail, and an untested verification step presented as part of the method), and seven editorial items. What survives: coverage in every experiment, the failure of static weights on a diverse family, the repair by nearest-anchor weights (confirmed on disjoint tuning data), and the robustness of field-space regions to both attacks.

1 Scope, conventions and method

Scope. The note safe_prompt_fields.tex as of 14 October 2026, 10:10 (1954 lines, 29 pages), archived unchanged with its number macros, its README and the three figure scripts that the revision modifies in archive/2026-10-14_pre_audit/. Line numbers (l. n) refer to that archived file; result, table and figure numbers to its compiled PDF. Experiments: E1 = Section 9 (pool of e1_fit.py, 300 safe and 400 unsafe prompts), E2 = Section 10 (pool of e2_pool.py). "1-rectified" and "reflowed" are the two models of E1; $P(A)$ is the fraction of outputs in the unsafe disk $A = \{\|\mathbf{y}\| \le 0.6\}$.

Severity scale.

- **[C: CHANGES A CONCLUSION]** — a conclusion of the note does not hold once the defect is removed.
- **[F: FALSE AS STATED]** — the printed statement admits a counterexample.
- **[O: OVERSTATED]** — the claim goes beyond its evidence or rests on a wrong mechanism, but its direction may be right.
- **[R: REPORTING]** — a reported number, check or protocol does not show what the text says it shows.
- **[S: SCOPE]** — a limitation that the note does not state.
- **[E: EDITORIAL]** — wording or bookkeeping; no change of content.
- **[CONFIRMED]** — statement and proof (or number and its script) verified, no action needed.

Method. *(i) Mathematics*: every proof of Sections 2–7 and Appendix A re-derived, including the measure-theoretic details (versions of conditional expectations, null sets at $t = 0$, well-posedness), and the cited theorem numbers checked against the bibliography's notes. *(ii) Code*: every script of E1 and E2 read against the protocol stated in the text, including the random-number streams, the noise sets used on each side of each inequality, and the pools used for tuning and evaluation. *(iii) Numbers*: the quoted numbers re-derived from `results/`; where a conclusion looked fragile, a targeted check was designed and run (Section 9).

2 Verdict table

Codes as in the severity scale: **[C]** changes a conclusion, **[F]** false as stated, **[O]** overstated, **[R]** reporting, **[S]** scope, [E] editorial, **[OK]** confirmed.

Id	Location	Code	Summary
Findings that change a conclusion			
F1	§9 Steps 4–5, Fig. 8, §10.1	**[C]**	One probe noise (of 256), sent beyond radius 4 by the 1-rectified sampler, produces "L^p, $p > 2$, fails" and "L^2 is less sensitive than truncated L^1".
F2	§10.4, Table 8	**[C]**	The pull-back region of the 1-rectified model admits prompts with $P(A) = 0.22$ (adaptive) and 0.85 (static); reported: 0.030, 0.262.
F3	abstract, §9 Step 8, §10.3, Q2	**[C]**	A Mahalanobis prompt region excludes every unsafe request and resists the attack; "fields, not prompts" is shown only against Euclidean prompt distances.
Mathematics			
M1	l. 1553, l. 1880–1881	**[F]**	"One-step models make [the improved region] convex": a union of balls, and $\min(\|\cdot\|, r)$ is not convex.
M2	abstract l. 108, l. 1084–1085, l. 1552, l. 1871	**[F]**	"Needs no Markov step": the proof of Proposition 7.5 is Markov's inequality applied to $\min(\delta, r)$.
M3	l. 149–150, l. 379–380	**[O]**	"e^L is about 10^{15}": 10^{15} is a discrete product with an a-priori bound $L \approx 52$; $e^{52} \approx 10^{22.6}$.
M4	l. 1350–1353, l. 1686–1687, l. 1807	**[O]**	Detection failure attributed to the floor of Proposition 5.6(iii); the budget is 2.5 times the floor.
M5	l. 1043–1048	**[O]**	"The L^2 map metric is the most sensitive" compares lower bounds across distances whose thresholds differ.
M6	l. 645, l. 655–656, l. 1389–1394, l. 1551	[E]	"Fidelity can only be lost"; Lemma 3.9(iii) applied to trimmed pairs; affine combinations called mixtures; "order-1 structure" kept in name only.
Methodology and reporting			
R1	l. 1359, Table 4, Fig. 8	**[R]**	"Coverage between 0.897 and 0.907 for every distance": the tail score has 0.918 and 0.934 (ties).
R2	Table 5, l. 1438	**[R]**	"No enforced field violates its certificate": both sides are measured on the same noise; the check cannot fail at this slack.
R3	l. 1440–1445, l. 1545–1551	**[R]**	The fresh-noise verification is part of "the improved method" but was never run; probability qualifier missing; Hoeffding margin eats 58% of the tail budget.
R4	abstract l. 108–109	**[O]**	"Field-space regions cap the unsafe mass at a few percent": the region as written admits 0.186 (E1) and 0.220 (E2).

Id	Location	Code	Summary
R5	l. 1859–1860	[R]	Q3 omits that the reflowed one-step model puts 0.0176 of approved outputs in A (7× the 1-rectified model).
R6	l. 1500–1501, l. 1671–1672, l. 1817–1818	[O]	Small-mass detection limits are confounded: the model realises only 24% ($m = 0.01$) and 40% ($m = 0.03$) of the requested unsafe mass.
R7	§9, §10	[R]	No uncertainty is reported; attacks on one split; E1's variants selected on the pool they are reported on.
R8	§9–§11	[S]	Nearest-anchor detection depends on the safe family being two-dimensional; not discussed.
R9	§7, §9, title	[S]	The improved method is a nearest-neighbour conformal novelty detector; related work absent.
R10	§11 l. 1873	[R]	"Tested: prompt mixing": only the algebraic identity was checked; the projection's results were computed and not reported.
Editorial and bookkeeping			
H1–H7	README, `e2_figures.py`, Table 2, l. 11	[E]	Stale page count, two numbers computed on the wrong pools, an unexplained coincidence in Table 2, a stale header comment, unused macros, a non-i.i.d. Clopper–Pearson bound.
Confirmed			
OK	§2–§7, App. A, all numbered results	[OK]	Statements and proofs correct; details in Section 7.

3 Findings that change a conclusion

F1 One probe noise drives the comparison of map distances **[C: CHANGES A CONCLUSION]**

What the note says. Section 9, Step 4 (l. 1322–1327): "the L^p distances with $p > 2$ [...] fail: high powers weigh the largest displacements, and between safe prompts the largest displacements are blob changes, which exceed the displacement of an unsafe mode. With uniform weights they exclude only 0.03 ($p = 4$) and 0.02 ($p = 8$) of the prompts with $m = 0.3$." Step 5 (l. 1359–1362): exclusion at $m = 0.1$ rises from 0.24 to 0.73 with the nearest field in L^2, but to 0.98 in L^1 and 0.99 in truncated L^1. Figure 8 caption (l. 1382–1383): the nearest field works "with L^2 (orange) only on the reflowed model". Section 10.1 (l. 1591–1592): the field region uses "the truncated map distance, the most sensitive field distance of Section 9".

What the data show.

- The 50-step Euler sampler of the 1-rectified model sends probe 136 ($|\mathbf{x}_0| = 3.96$, the largest of the 256 frozen probes) beyond radius 4 for 44% of the safe prompts, up to $|T| = 16.5$; the ring has radius about 2. No other probe leaves radius 4 for any pool prompt, in E1 or in E2, and the same probe is the only one flagged for the reflowed model ($|T| \leq 7.6$). On the evaluation

noise of E1 the 1-rectified sampler reaches $|T| = 2.8 \times 10^4$; on the probes, for the prompts of the E2 admission grid (rotations up to $\pm 45°$, outside the approved range), 2.1×10^7; and its reflow pairs reached 2.5×10^5 before trimming. The note mentions this instability only to justify the trimming of the reflow pairs (l. 652–653).

- Between two safe prompts, probe 136 alone carries a median 6% (mean 19%, 90th percentile 57%) of the squared L^2 map distance, and a median 35% of the sum that defines the L^4 distance.
- The unsafe mode suppresses the blow-up: probe 136 leaves radius 4 for 44% of the safe prompts but 55%, 57%, 34% and 3% of the prompts with $m = 0.01, 0.03, 0.1, 0.3$. Hence the inverted dose-response of the uniform L^4 region, which excludes 0.17 of the requests with $m = 0.01$ and 0.03 of those with $m = 0.3$: the signature of a nuisance variable, not of a less sensitive distance.
- Removing that one probe (Table 2) removes every effect the note attributes to the distance: uniform L^4 and L^8 exclude 1.00 and 1.00 at $m = 0.3$; the nearest-field L^2 region excludes 1.00 at $m = 0.1$ on the 1-rectified model, as L^1 and truncated L^1 do; in E2, nearest-field L^2 excludes 1.00 at $m = 0.1$ and 0.22 at $m = 0.03$, against 0.20 for truncated L^1. The bounded distances barely move.

Consequence. The mechanism of Step 4 is wrong, and the claimed reversal of Proposition 7.3 ("L^2 is not the best metric") does not exist: once the outlier is removed, L^2 is at least as sensitive as truncated L^1. Truncation is still useful here, but as protection against a sampler that diverges on rare noise, which the note does not discuss.

Fix for F1

State the sampler's instability where the models are introduced; replace the mechanism of Step 4 by the probe diagnosis; justify the truncated distance by its boundedness (robustness to diverging trajectories, and a certificate with a bounded variable), not by sensitivity; report Table 2; correct the caption of Figure 8 and Section 10.1. A cleaner long-term repair is a sampler that cannot diverge (clipping, or features with bounded extrapolation).

F2 The pull-back region of the 1-rectified model is attackable **[C: CHANGES A CONCLUSION]**

What the note says. Section 10.4 (l. 1832–1833): the pull-back metric "with adaptive weights [...] caps the attack at a few percent on both models"; Table 8 reports the worst attack $P(A)$ falling from 0.976 (Euclidean prompt region) to 0.030 (pull-back, 1-rectified, adaptive).

What the data show. The pull-back Gram matrix $Q = U^\top G_0 U$ of the 1-rectified model has eigenvalues from 1.5×10^{-3} to 2.4×10^4 (condition number 1.6×10^7). The note's attack proposes Gaussian steps in all 43 coordinates of the embedding, so nearly every proposal moves along a stiff direction and is rejected. *Attack 2* searches instead the five eigen-directions of the region's metric with the smallest eigenvalues (for a field distance: of $J^\top J$, J the Jacobian of the probe outputs), from the first twelve admitted approved prompts of the same split, 400 steps each, and reports $P(A)$ on fresh noise. It finds admitted embeddings with $P(A) = 0.22$ (adaptive) and 0.85 (static) inside the pull-back region of the 1-rectified model (Table 3). It does not break the field regions (worst 0.071 and 0.019 for the 1-rectified model, 0.047 and 0.029 for the reflowed one), nor the pull-back region of the reflowed one-step model (0.049 and 0.024), whose pull-back

Table 2: Probe sensitivity (F1): exclusion of unsafe requests and median budget with all 256 probes → without probe 136, the only probe that some pool prompt's sampler sends beyond the trimming radius 4. Same splits, scores and thresholds as the note; coverage stays within $[0.897, 0.934]$ in every row.

	excluded at $m =$				
score	0.01	0.03	0.1	0.3	budget
E1, 1-rectified model (100 splits)					
uniform, L^2	0.09→0.07	0.07→0.08	0.24→0.33	1.00→1.00	0.829→0.739
uniform, L^4	0.17→0.07	0.17→0.07	0.08→0.30	0.03→1.00	1.816→0.911
uniform, L^8	0.18→0.08	0.17→0.08	0.08→0.18	0.02→1.00	3.522→1.163
uniform, truncated L^1	0.08→0.08	0.09→0.09	0.31→0.31	0.98→0.98	0.598→0.597
nearest, L^1	0.11→0.10	0.19→0.18	0.98→0.98	1.00→1.00	0.192→0.178
nearest, L^2	0.13→0.11	0.17→0.20	0.73→1.00	1.00→1.00	0.377→0.239
nearest, L^4	0.14→0.09	0.14→0.15	0.40→1.00	0.89→1.00	0.660→0.389
nearest, truncated L^1	0.10→0.11	0.19→0.19	0.99→0.99	1.00→1.00	0.178→0.176
nearest, tail score	0.10→0.10	0.29→0.29	1.00→1.00	1.00→1.00	0.043→0.039
E1, 2-rectified model (100 splits)					
uniform, L^4	0.10→0.08	0.10→0.09	0.17→0.30	0.97→1.00	0.967→0.769
uniform, L^8	0.10→0.08	0.08→0.09	0.07→0.25	0.23→1.00	1.715→0.954
nearest, L^2	0.10→0.11	0.19→0.19	1.00→1.00	1.00→1.00	0.188→0.178
nearest, truncated L^1	0.10→0.10	0.18→0.18	0.98→0.98	1.00→1.00	0.156→0.155
E2, field regions of the 1-rectified model (evaluation pool, 200 splits)					
uniform, L^2	0.08→0.13	0.08→0.14	0.25→0.39	0.99→1.00	0.842→0.733
uniform, truncated L^1	0.12→0.12	0.13→0.13	0.35→0.35	0.98→0.98	0.593→0.592
nearest, L^2	0.09→0.11	0.13→0.22	0.74→1.00	1.00→1.00	0.374→0.229
nearest, truncated L^1	0.11→0.11	0.20→0.20	0.99→0.99	1.00→1.00	0.172→0.170

metric is its field metric; on the E1 regions its worst value is 0.084.

Why. This is what Lemma 4.3(iii) predicts: the exact field at $t = 0$ depends on the target only through its mean, so a metric on the slice at $t = 0$ cannot constrain the outputs of a curved 50-step sampler. Low-eigenvalue directions of Q barely move the slice at $t = 0$ and move the later times freely.

Table 3: Worst $P(A)$ (fresh noise) found inside each region by the note's attack (12 random-search runs over all embedding coordinates, from enforced requests with $m = 0.3$) and by attack 2 (12 runs in the five directions the region penalises least, from admitted approved prompts). The Euclidean metric has no preferred direction, so attack 2 is not defined for it.

	note's attack	attack 2	
region	worst	worst	median
E1, 1-rectified model, split 0			
field, uniform L^2 (as written)	0.186	0.084	0.046
field, nearest L^1	0.054	0.016	0.006
field, nearest truncated L^1	0.051	0.018	0.007
field, nearest tail score	0.041	0.024	0.007
E1, 2-rectified model, split 0			
field, uniform L^2 (as written)	0.099	0.056	0.017
field, nearest L^1	0.041	0.024	0.011
field, nearest truncated L^1	0.036	0.024	0.011
field, nearest tail score	0.044	0.020	0.012
E2, 1-rectified model (50 steps), evaluation split 0			
prompt, Euclidean, static	1.000	–	–
prompt, Euclidean, adaptive	0.976	–	–
prompt, Mahalanobis, static	0.003	0.026	0.013
prompt, Mahalanobis, adaptive	0.004	0.044	0.017
pull-back, static	0.262	0.854	0.559
pull-back, adaptive	0.030	0.219	0.114
field, truncated L^1, static	0.220	0.071	0.051
field, truncated L^1, adaptive	0.036	0.019	0.006
E2, reflowed model (one step), evaluation split 0			
prompt, Euclidean, static	0.655	–	–
prompt, Euclidean, adaptive	0.496	–	–
prompt, Mahalanobis, static	0.016	0.042	0.030
prompt, Mahalanobis, adaptive	0.023	0.056	0.031
pull-back, static	0.099	0.049	0.027
pull-back, adaptive	0.030	0.024	0.020
field, truncated L^1, static	0.072	0.047	0.027
field, truncated L^1, adaptive	0.039	0.029	0.021

Fix for F2

Restrict the robustness claim of the pull-back metric to the reflowed one-step model, report attack 2 for every region, and state that a pull-back of the slice at $t = 0$ of a curved model is not a safety device.

F3 The prompt-space baseline is the weakest one available **[C: CHANGES A CONCLUSION]**

What the note says. Abstract (l. 108–109) and Section 9, Step 8 (l. 1476–1477): "constraining fields rather than prompts is what protects against adversarial prompts"; Section 10.3 (l. 1771–1773): a prompt region "fails, static or adaptive, under attacks on the embedding or wording that the model ignores. A field region resists both."

What the data show. Every prompt region of the note uses the Euclidean distance between embeddings. Replace it by the Mahalanobis distance with the covariance S of the approved embeddings of the *tuning* pool, regularised by $\rho\,\mathrm{tr}(S)/43$, and tune ρ and the adaptive weights on the tuning pool exactly as the note tunes its other regions ($\rho = 1 \times 10^{-6}$; adaptive weights: Gaussian kernel with $c = 0.03$). On the evaluation pool:

- coverage 0.905 (static) and 0.905 (adaptive); exclusion 1.00, 1.00, 1.00, 1.00 at $m = 0.01, 0.03, 0.1, 0.3$ (static) — including $m = 0.01$, of which no field region excludes more than 0.24 even with 240 anchors;
- on the 1-rectified model, the note's attack finds at most $P(A) = 0.003$ (static) and 0.004 (adaptive), against 0.036–0.220 for the field regions; attack 2 at most 0.026 and 0.044;
- its back-off is crude: requests with $m = 0.3$ end at $P(A) = 0.001$ but with $\mathrm{MMD}^2 \times 10^3 = 91$ (static) and 101 (adaptive) to the request's safe target, against 11 and 26 for the adaptive Euclidean prompt and field regions;
- with the nuisance token of Section 10.3, at every scale λ, exclusion at $m = 0.1$ stays at least 1.00.

Why. The approved embeddings of the toy are degenerate: their third coordinate $m/0.2$ is identically 0 (variance 0 on the tuning pool) and they lie on a two-dimensional surface, parametrised by rotation and radius, of $\mathbb{R}^{43}$. Any detector that whitens by the approved covariance flags every unsafe request ($m > 0$) and every embedding that leaves the surface, which is what every attack does. The comparison is therefore between field regions and the weakest prompt region, not between fields and prompts.

Fix for F3

Report the Mahalanobis baseline; restrict the conclusion to Euclidean prompt distances; state that the toy cannot separate prompt-space from field-space regions beyond that, because its approved embeddings are degenerate and its encoder exposes m. A meaningful comparison needs approved embeddings that span their space and an encoder that hides the unsafe content.

4 Mathematics

M1 "One-step models make [the improved region] convex" (l. 1553) **[F: FALSE AS STATED]**

The improved region is $\{\boldsymbol{\theta} : \min_i d_r(\boldsymbol{\theta}, \boldsymbol{\theta}_i) \le (M+1)\varepsilon\} = \bigcup_i \{\boldsymbol{\theta} : d_r(\boldsymbol{\theta}, \boldsymbol{\theta}_i) \le (M+1)\varepsilon\}$, a union of balls around the anchors, which is not convex for any sampler as soon as two balls are disjoint. Each ball is not convex either: for a one-step sampler, $\boldsymbol{\theta} \mapsto T_{\boldsymbol{\theta}}(\mathbf{x})$ is affine, but $\min(\|\cdot\|, r)$ is not convex and its average over $\mathbf{x}$ has non-convex sublevel sets.

Counterexample

One dimension, two noise points with equal weight, $T_\theta(x_1) = \theta$, $T_\theta(x_2) = \theta - 10$, anchor $\theta_1 = 0$, $r = 1$: $f(\theta) = \frac{1}{2}\min(|\theta|, 1) + \frac{1}{2}\min(|\theta - 10|, 1)$ gives $f(0) = f(10) = 0.5$ and $f(5) = 1$. The ball $\{f \le 0.5\}$ contains 0 and 10 but not 5.

The same sentence reappears in Section 11 (l. 1880–1881: "one-step samplers resolve [the tension between convexity and safety]") and contradicts Section 10.4 (l. 1818–1820), which correctly calls the adaptive region non-convex. Convexity holds for one-step samplers only with static weights and a Mahalanobis distance (Corollary 7.2(i)), the configuration the experiments show to fail. **Fix:** delete the claim; state that the improved region is non-convex and enforced by back-off.

M2 "A truncated map metric whose certificate needs no Markov step" **[F: FALSE AS STATED]**

Abstract l. 108; "the factor $1/r$ of Theorem 7.1(ii) is gone" (l. 1084–1085); "the absence of a Markov factor" (l. 1552); "without Markov factor" (l. 1871). The proof of Proposition 7.5 rests on $\mathbb{1}\{\delta \ge r\} \le \min(\delta, r)/r$, i.e. on $\mathbb{P}(Z \ge r) \le \mathbb{E}Z/r$ for $Z = \min(\|Y - Y'\|, r)$, since $\{Z \ge r\} = \{\|Y - Y'\| \ge r\}$: Markov's inequality applied to the truncated displacement. The factor $1/r$ has moved into the normalisation of d_r; it is not gone. What truncation buys is real but different: $d_r \le d_{L^1}/r$, because displacements beyond r are not charged, so the certificate is never worse than the L^1 one and is much better when large displacements are present. Only the tail score $\mathbb{P}(\|T_{\boldsymbol{\theta}} - T_{\boldsymbol{\theta}'}\| \ge r)$ avoids Markov's inequality, and it is not a metric. **Fix:** "Markov's inequality on a bounded variable, which does not charge displacements beyond r".

M3 The size of the Grönwall factor (l. 149–150, l. 379–380) **[O: OVERSTATED]**

Remark 2.14 derives $\|X_1 - X_1'\| \le (\|X_0 - X_0'\| + b)e^L$ and says that this factor "is about 10^{15} in the toy". The 10^{15} comes from the proof-of-concept note: it is the discrete product $\prod_{l=1}^{J-1}(1 + hL_l)$ with an *a-priori* bound $L_l \in [52, 53]$, obtained from the triangle inequality over the features (`make_tables.py`, Grönwall block). With $L = 52$, $e^L \approx 10^{22.6}$; and either number bounds the factor from above, through a bound on the Lipschitz constant that may be far from tight. The conclusion (the Grönwall route is vacuous here) is plausible but not established by this number. **Fix:** "an a-priori bound on this factor is about 10^{15}".

M4 The floor of Proposition 5.6(iii) is not the mechanism of the detection failure **[O: OVERSTATED]**

Section 9, Step 5 (l. 1350–1353): the unsafe request's score stays below the threshold, and "this is the floor of Proposition 5.6(iii)"; likewise l. 1686–1687 and l. 1807. The floor bounds the *smallest*

score, half the mean distance between anchors. Detection depends on the margin between the threshold and the scores of typical prompts. On the 1-rectified model (100 splits) the budget $(M+1)\varepsilon$ is 0.83, the floor 0.33 (ratio 2.5), and a typical safe test prompt has a mean distance 0.68 to the anchors; the margin $0.83 - 0.68$ is the spread of the safe scores, set by rotation. For L^2, a far anchor also adds the unsafe displacement in quadrature, $d(u, \boldsymbol{\theta}_i) \approx (d_i^2 + \delta^2)^{1/2}$, which is why averaging dilutes it. **Fix:** attribute the failure to the spread of the safe scores; keep the floor as a statement about the tightness of the certificate.

M5 "Of the three, the L^2 map metric is the most sensitive" (l. 1043–1048) [O: OVERSTATED]

Proposition 7.3 is correct. The sentence after it compares the guaranteed lower bounds $\sqrt{m}\Delta$ (L^2), $m\Delta$ (L^1) and $\log \frac{1}{1-m}$ (forward divergence), but a request is excluded when its score exceeds a threshold that is calibrated separately for each distance and grows with it. Sensitivity is a ratio between the signal and the spread of the safe scores. After F1 the claim happens to hold in the experiments, but the argument does not establish it. **Fix:** compare the lower bound with the calibrated budget of the same distance.

M6 Minor mathematical wording [E: EDITORIAL]

- l. 645: "fidelity can only be lost" by reflow. Not a theorem: the exact reflowed flow reproduces ν_v, and a refit error can move the outputs towards p as well as away. Here fidelity drops ($\mathrm{MMD}^2 \times 10^3$ from 3.3 to 8.1).
- l. 655–656: Lemma 3.9(iii) is applied to "the exact reflowed field" of E1 because "its sampler map is Lipschitz". True, but the constant is so large (the sampler sends rare noises to 2.5×10^5) that the interval $t < 1/(1+L)$ on which the lemma describes the field is negligible; what is used is the identity at $t = 0$, which survives. The trimming of the pairs with $|T(\mathbf{x}_0)| > 4$ is harmless: the pairs are $(\mathrm{id}, T)_{\#}\mu_0'$ for the conditioned noise law μ_0', and the proof of (iii) applies to them verbatim. The statement should say both.
- l. 1389–1394: the back-off of Step 6 is the field of a mixed prompt because the model is affine in the prompt (Lemma 2.19(i)), not by Corollary 6.3, which concerns the projection; and the plane of Figure 9 contains affine combinations with negative weights, not only mixtures.
- l. 1551: "this keeps [...] the order-1 Wasserstein structure". With a Dirac weight on the candidate's nearest anchor the reference law depends on the candidate, and the score is the nearest-anchor distance; the region is the union of balls that the note itself compares with the order-∞ region (l. 1083–1084). The Wasserstein reading is literal but carries no content (see R9).

5 Methodology and reporting

R1 Coverage is misreported for the tail score [R: REPORTING]

Step 5 (l. 1359): "coverage lies between 0.897 and 0.907 for every distance and model". These are the uniform scores of phase 1. Figure 8, which the sentence cites, also shows the nearest-field scores; over the same 100 splits their coverage ranges from 0.897 to 0.934, and the tail score reaches 0.918 (1-rectified) and 0.934 (reflowed). The macro `\eoneKnnTailCov` (= 0.918) is

computed by `e1_tables.py` and never printed. The cause is ties: the tail score counts probes, and its median threshold is 11 probes out of 256 on the 1-rectified model and 2 on the reflowed one (with $r = 1$ on the reflowed model the threshold is 0.000 and coverage 0.958). Over-coverage is valid, but the region is then decided by a handful of probes, and the "tightest certificate" of the note rests on that budget. **Fix:** report the range of every score shown; explain the ties; randomise ties or use more probes.

R2 "No enforced field violates its certificate on fresh noise" cannot fail [R: REPORTING]

Table 5 and l. 1438. In `e1_phase2.py` (`certify`, lines 96–119, and `run_method`, line 135) the term $q_i(A^{+r})$ of the bound and the reported $P(A)$ are both measured on the same 2000 evaluation noises. On that sample, $\mathbb{1}\{Y \in A\} \le \mathbb{1}\{Y' \in A^{+r}\} + \mathbb{1}\{\|Y - Y'\| \ge r\}$ holds noise by noise, so the empirical inequality can only fail if the fresh distance exceeds the probe budget by more than the slack of the bound; for the median enforced field of the improved method the bound is 0.300 for $P(A) = 0.030$ (tail score: 0.086 for 0.026). The "0/2400" therefore restates that no fresh distance exceeds its budget by an order of magnitude, which the next column already reports (at most 1.13 times). **Fix:** report the slack and drop the claim, or test the bound on noise independent of every term (Section 9).

R3 The verification step of the improved method was never run [R: REPORTING]

Step 7 (l. 1441–1445) proposes to verify each enforced field on L fresh probes with Hoeffding's inequality, and "The improved method" (l. 1545–1551) lists that verification as a step and concludes that "the deployed outputs then satisfy" the certificate. No script implements it; the deployed fields of Tables 4–5 are plug-in, and 8%–28% of them exceed their budget on fresh noise. The conclusion also omits the probability: with the verification it holds with probability at least $1 - \delta$. Finally, the margin matters: with $L = 2000$ and $\delta = 0.05$, Hoeffding adds 0.027, i.e. 58% of the median tail budget (0.047) and 15% of the truncated one (0.182). For the tail score, a proportion, the Clopper–Pearson bound is much tighter (no event in 2000 gives 0.0015). **Fix:** implement and evaluate the verification, with Clopper–Pearson for the tail score; until then, present it as untested.

R4 "Field-space regions cap the unsafe mass at a few percent" (abstract, l. 108–109) [O: OVERSTATED]

The field-space region as written admits prompts with $P(A) = 0.186$ (E1, Table 5) and 0.220 (E2, static); only the localised regions stay at a few percent (0.054 and below). The evidence is twelve random-search runs on a single split, a lower bound on the worst case; the bounds that hold for every field of the region are 0.34 (truncated) and 0.09 (tail), and they are plug-in. **Fix:** "localised field-space regions", and state what the evidence is.

R5 The recommended configuration hides a safety cost (l. 1859–1860) [R: REPORTING]

Q3 recommends the reflowed model sampled in one step, "at the fidelity cost of one-step sampling". That model puts $P(A) = 0.0176$ of the outputs of *approved* prompts in A, 7 times the 1-rectified model (0.0024); this is comparable to what the adaptive regions of the 1-rectified model leave after enforcing requests with $m = 0.3$ (0.009–0.026). The number is computed

(`\eoneOneStepPA`) and not reported. **Fix:** report it next to the fidelity cost.

R6 Small-mass detection limits are confounded by under-generation [O: OVERSTATED]

The note reads the low exclusion at $m = 0.01$ and $m = 0.03$ as limits of the regions (l. 1500–1501, l. 1671–1672, l. 1817–1818). The 1-rectified model realises an excess $P(A)$ over approved prompts of 0.0023 at $m = 0.01$ and 0.0112 at $m = 0.03$, i.e. 24% and 40% of the targets' 0.0095 and 0.0284: these fields are much closer to safe fields than their nominal mass suggests. **Fix:** report exclusion against the realised excess $P(A)$, or state the realised masses.

R7 Statistical precision [R: REPORTING]

No interval accompanies any number of Sections 9–10. Exclusion at a mass is a mean over 100 distinct requests (E1 and E2), so a binomial 95% half-width is up to ±0.10 whatever the number of splits; enforcement uses 5 (E1) and 3 (E2) splits; every attack uses one split and twelve runs. In E1, the improved variants (nearest field, $r = 1$, tail at 0.5) were selected by a screen on the same pool as the numbers reported for them; E2 repairs this with disjoint tuning and evaluation pools and confirms the selection. **Fix:** state the precision and the selection.

R8 External validity of nearest-anchor detection [S: SCOPE]

The approved family is two-dimensional (rotation, radius). The distance from a request to its nearest anchor shrinks like $M^{-1/d}$ for a d-dimensional family, and already here $m = 0.03$ needs 240 anchors (Figure 16). Nothing in the note discusses what happens when the approved prompts span many dimensions. **Fix:** state the dependence on the intrinsic dimension.

R9 Framing and related work [S: SCOPE]

With weights on the nearest anchor, the region is the split-conformal nearest-neighbour novelty detector (Laxhammar and Falkman, *Ann. Math. Artif. Intell.* 74, 2015; Bates et al., *Ann. Statist.* 51, 2023), applied to a distance between sampler maps; the experiments favour it over the order-1 score that gives the note its title. Prompt-space regions with a whitened covariance are a standard out-of-distribution detector (Lee et al., NeurIPS 2018). None is cited. **Fix:** say so where the improved method is stated, and cite.

R10 "Tested: [. . .] prompt mixing" (l. 1872–1874) [R: REPORTING]

What E1 checks is the identity of Corollary 6.3 (largest deviation 5.7×10^{-13} in `eval_enforce.json`), which holds by linearity. The projection of Proposition 6.2(iii), the constructive result of Section 6, was computed in phase 1 (`e1_eval.py`) and its effect on $P(A)$ is never reported; every enforcement in the note is a back-off. **Fix:** say what was checked, or report the projection.

6 Editorial and bookkeeping

H1 `README.md` calls the note 23 pages; it has 29.

H2 `e2_figures.py`, lines 325–327: the spread `\etSpreadOne` averages over the evaluation prompts against *all* 1000 approved prompts, diagonal zeros included, and the displacement `\etDispOne`

uses the 200 bases of both pools and the nearest of all 1000 approved prompts, whereas the text describes evaluation-pool quantities. The values barely change (spread 0.485 against 0.484 between distinct evaluation prompts; displacement 0.236 against 0.238).

H3 Q1 (l. 1854–1856) speaks of weights "that take the transport mass from the nearest anchors"; the adaptive field region of the reflowed model, tuned on the tuning pool, uses prompt-kernel weights (Table 7: "$c = 0.3$, prompt").

H4 Table 2: the rows "G_{on}, ODE" and "reverse KL, SDE" agree to two decimals in all five columns. This is a coincidence of rounding, not a copy error: at $m = 0.01$ and $m = 0.03$ the exclusions are 0.133, 0.552 and 0.129, 0.546. A caption note avoids the question.

H5 l. 11: "Every number in Section 7 is copied from `tables/numbers.tex`" — that section is now Section 8, and its Table 2 is typed by hand.

H6 Computed but never printed: `\eoneKnnTailCov`, `\eonePlugFail`, `\eonePlugMax`, `\eoneRegionWorst`, `\eoneOneStepPA`, `\etTuneGap`, `\etPoolPATwo`; R1 and R5 show that two of them change the reading of the results.

H7 Section 8: the Clopper–Pearson bound on $\bar{P}_w(A)$ (`part_d.py`, line 82) treats the $M \times 4000$ indicators as independent, but the M anchors share the same 4000 noise chains; the bound is slightly anti-conservative.

7 What was checked and holds

[CONFIRMED] Section 2. Definitions 2.1, 2.3, 2.7, 2.9, 2.11, 2.15, 2.17, 2.20, 2.21; Lemma 2.4 (including (iii): truncation of a metric is a metric, and (iv): integrals of pseudometrics); Lemma 2.5; Theorem 2.6 (i)–(iii); Proposition 2.8; Lemma 2.10, including Bayes' formula with the null set where the ratio is undefined; Lemma 2.13; the Grönwall inequality of Remark 2.14; Lemma 2.16; Lemma 2.19.
[CONFIRMED] Section 3. Remark 3.2, including "for almost every s" from conditioning on $(\mathbf{x}_t, t)$ jointly; Lemmas 3.3 and 3.6; Lemma 3.9: (ii) through the dyadic-cube argument and two versions of $\mathbb{E}[\mathbb{1}_B(\mathbf{x}_1) \mid \mathbf{x}_t, t]$, (iii) through the contraction $\mathbf{x} \mapsto (\mathbf{y} - tT(\mathbf{x}))/(1-t)$, (iv) through strong monotonicity of Ψ_t and the induction on the Euler iterates; the identity of Liu et al. behind Remark 3.10, re-derived (crossing plus curvature) and checked against the figure's numbers (5.21 against 5.25).
[CONFIRMED] Sections 4–7. Proposition 4.2; Lemma 4.3 (the Bayes version is continuous on $\mathbb{R}^n \times [0,1)$ and equals $\mathbb{E}_p[\mathbf{x}_1] - \mathbf{x}$ at $t = 0$); Lemma 5.3 (plan and dual certificate); Proposition 5.4 (the rank argument holds conditionally on $\mathcal{D}$ in the sense of Definition 2.21); Proposition 5.6; Proposition 6.2, including the convex-hull argument of (iii); Corollary 6.3; Proposition 6.4; Theorem 7.1; Corollary 7.2; Proposition 7.3; Proposition 7.5 (the inequality $\mathbb{1}\{\delta \geq r'\} \leq \max(1, r/r') \min(\delta, r)/r$ in both cases $r' \leq r$ and $r' > r$).
[CONFIRMED] Appendix A. The drift and diffusion of Definition A.1 match the marginal-preserving SDE of the linear interpolant; Lemma A.2 (chain rule over Gaussian kernels); Proposition A.3 (i)–(iii).
[CONFIRMED] Experiments. Split-conformal thresholds implement $C_{1-\beta}$ exactly (`kth_threshold`); every reported coverage is within sampling error of 55/61 or 37/41 (or above it, R1); tuning and evaluation pools of E2 are disjoint, and the tuned weights are within 0.005 of the best on the evaluation pool; the pull-back metric of the reflowed model equals its one-step map metric to 7×10^{-8}; the attack reports $P(A)$ on noise not used by the search; static regions are too

wide on this family (E1 and E2), and nearest-anchor weights repair detection; field regions resist both attacks (Table 3).

8 Resolution in the revised note

The note was revised on 14 October 2026, in place (`safe_prompt_fields.tex`, now 33 pages); the audited version is archived. Section, table and figure numbers below refer to the revised note.

Id	Resolution
F1	The sampler's divergence is stated where the models are introduced (Section 9, Step 1; Remark 3.10). Step 4 replaces the "blob changes" mechanism by the probe diagnosis, with a table of every score with and without the flagged probe (Table 4); Step 5 and the caption of Figure 8 (which now shows nearest-field L^2 without the probe) give L^2 its corrected standing; Section 10.1 justifies the truncated distance by its boundedness. The paragraph after Proposition 7.3 warns that unbounded distances inherit far-away trajectories.
F2	The directed search (attack 2) is part of both experiments: Section 9, Step 8, and its certificate table (Table 7), Table 9 and Figure 14 of Section 10, and the pull-back rows of Table 10. Section 10.4 states that the pull-back metric of the 1-rectified model is not a safety device, with the directed-search numbers; the summary (Q3) and Section 11 say so too.
F3	The Mahalanobis prompt region is defined in Section 10.1 (Table 8), evaluated in Table 9, Figures 14–15 and Table 10, and discussed in a new paragraph of Section 10.3, including its crude enforcement. The abstract, Step 8, the take-away of Section 10.3, Q2 and Section 11 restrict the conclusion to Euclidean prompt distances and state why the toy cannot separate the two spaces.
M1	The convexity claim is deleted ("The improved method", Section 11); after Proposition 7.5 the union-of-balls region is stated to be non-convex for any sampler.
M2	Every "no Markov step" is replaced by "Markov's inequality on a bounded variable", and the real gain, $d_r \le d_{L^1}/r$, is stated after Proposition 7.5.
M3	Remark 2.14 and Section 1 speak of an a-priori bound on the factor, quoting the proof of concept's own macros.
M4	Section 9, Step 5, and Sections 10.2 and 10.4 attribute the failure of static regions to the spread of the safe scores, with the budget, the typical score and the floor side by side.
M5	The paragraph after Proposition 7.3 compares the lower bounds with the calibrated budget of each distance.
M6	Remark 3.10 (fidelity; Lemma 3.9(iii) for the trimmed pairs and a large Lipschitz constant); Step 6 (Lemma 2.19(i) for the back-off, affine combinations in the plane of Figure 9); the "order-1 structure" sentence is gone.
R1	Step 5 gives the coverage of every score of Figure 8, the over-coverage of the tail score and its cause.
R2	The certificate table of Section 9 (Table 7) reports the median $P(A)$ next to the median bound instead of the violation count, and Step 7 explains why the count was uninformative.

Id	Resolution
R3	The verification is implemented (`code/e1_verify.py`) and reported in Step 7 with its own table (Table 5): a two-stage check on 2000 fresh noises, Hoeffding margins for the truncated distance and Clopper–Pearson for the tail score, fallback to the nearest anchor, and an evaluation on independent noise. On the 1-rectified model, the deployed fields of the improved region exceed their budget on 10,000 independent noises in 0.0% of cases (against 8%–28% on the evaluation noise before), and none of the 5600 deployed fields has $P(A)$ above its certificate on noise independent of every term (0 violations). "The improved method" carries the probability qualifier.
R4	The abstract speaks of localised field-space regions and two attacks.
R5	Section 10.4 and Q3 report the $P(A)$ of approved prompts under one-step sampling.
R6	Section 9, Step 1, gives the realised unsafe masses; the summaries of Sections 9 and 10.4 refer to them.
R7	Section 9, Step 1, and Section 10.1 state the precision of exclusion rates and that every attack uses one split.
R8	Section 10.4 and Section 11 state the dependence of nearest-anchor detection on the dimension of the approved family.
R9	After Proposition 7.5 and in "The improved method": the nearest-neighbour conformal novelty detector, with Laxhammar and Falkman (2015) and Bates et al. (2023); Lee et al. (2018) for the Mahalanobis baseline (`refs.bib`, Part 5, checked against the sources).
R10	Step 6 says what was checked (the identity of Corollary 6.3) and that every score is enforced by back-off; Section 11 lists "the identity of prompt mixing".
H1–H7	README page count and pipeline; `e2_figures.py` computes spread and displacement on the evaluation pool; Q1 names the prompt-kernel weights of the reflowed field region; the caption of Table 2 explains the coincidence (three-decimal macros from `make_tables.py`) and the shared-noise caveat; the header comment is corrected and Section 8 quotes the proof of concept's macros; the two unused macros that mattered (R1, R5) are printed.

9 Reproduction

From `code/`, after the pipelines of the README (E1 and E2):

script	content	time
`python e1_robustness.py`	F1 on E1 (screen with and without probe 136), attack 2 on the E1 regions	5 min
`python e2_robustness.py`	F1 on E2, attack 2 on the E2 regions, the Mahalanobis regions (F3)	15 min
`python e1_verify.py`	R2–R3: verified deployment, tested on independent noise	45 min
`python audit_checks.py`	every number and both tables of this audit	30 s

The audited note, its number macros and the figure scripts changed by the revision are archived in `archive/2026-10-14_pre_audit/`.